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Successful treatment and long-term follow-up of a young cat with feline infectious peritonitis and renal lymphoma

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ABSTRACT

Background: Feline lymphoma is often a fatal disease, and achieving a cure with chemotherapy is questionable. Feline infectious peritonitis (FIP) diagnosis was historically associated with a poor prognosis and high mortality, but remission is possible with the newly available treatment. Long-term follow-up data on treated patients remain limited.

Case description: A 3.9-year-old neutered male cat presented with hyporexia and lethargy lasting for several days. An abdominal mass was palpated on clinical examination and confirmed to be a renal mass on a computed tomography angiography scan. Ultrasound-guided fine needle aspirates and cytological evaluation revealed a large-cell renal lymphoma. A modified Cyclophosphamide, vincristine and prednisolone protocol with vinblastine addition (Cyclophosphamide, vincristine, prednisolone and vinblastine protocol) was started, resulting in a complete response. Twenty weeks after chemotherapy initiation, the cat developed anorexia, fever, and pleural effusion. FIP was diagnosed based on a positive reverse transcription polymerase chain reaction test from the pleural effusion. Subcutaneous administration of GS441524 led to FIP resolution.

Conclusion: The cat did not receive any further treatment for both FIP and lymphoma. Still alive and well, almost 5 years after the diagnosis. To the author's knowledge, this is the first case of renal lymphoma in a cat that subsequently developed FIP. Further studies on the potential correlation between lymphoma, chemotherapy, and FIP are needed.

Keywords: CHOP protocol, FIP, Remdesivir, Renal lymphoma.

Introduction

The pathogenesis of feline infectious peritonitis (FIP) is not yet entirely understood, but the virulent biotype Feline Infectious Peritonitis Virus (FIPV) is caused by a mutation in the feline enteric coronavirus (FCoV) with changes in cell tropism from enterocyte to monocyte (Müller *et al.*, 2023; Tasker *et al.*, 2023).

Immunohistochemistry is considered the gold standard for diagnosis (Felten and Hartmann, 2019); however, a combination of clinical signs and effusion analysis, if present, including cytology, biochemistry, and FCoV RNA polymerase chain reaction testing, would strongly support the diagnosis of FIP (Thayer *et al.*, 2022; Tasker *et al.*, 2023). Immune-mediated vasculitis, typical of the wet form, abdominal effusion, lymphadenopathy, and subcapsular kidney abnormalities effusion have been commonly reported in infected cats (Lewis and O'Brien, 2010; Müller *et al.*, 2023). Chest involvement with pathological lesions is less common, including pleuritis, vasculitis, perivascular parenchymal injury, and pneumonia, which can occur, leading to pleural

effusion and dyspnea without the typical systemic signs of FIP (Slaviero *et al.*, 2024). Abdominal effusions, lymphadenopathy, and pyogranulomatous abdominal lesions, the eyes, and the central nervous system are some common findings in the wet and dry (Kipar *et al.*, 2005; Tasker, 2018, 2023). Renal abnormalities are common in younger cats, including bilateral or unilateral renomegaly and cortical hyperechogenicity, medullary rim sign, pyelectasia, and corticomedullary differentiation loss (Kipar *et al.*, 2005; Tasker, 2018; Gülersoy *et al.*, 2023; Müller *et al.*, 2023). Poor prognosis and high fatality rate historically associated with FIP have improved considerably in recent years due to the novel use of nucleoside analogs, which have been successfully cleared and completed long-term resolution of clinical signs (Dickinson *et al.*, 2020; Coggins *et al.*, 2023; Green *et al.*, 2023). However, the long-term follow-up of these patients with this new drug remains limited.

Lymphoma is a hematopoietic neoplasia that occurs due to malignant cell proliferation of lymphocytes

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Table 1. Hematology when the cat was diagnosed with renal lymphoma. Abnormalities found were neutrophilia and monocytosis.

Blood parameter	Value patient	Normal values
RBC	7.37 x 10 ¹² /l	6.54–12.20
HCT	34.0%	30.3–52.3
HGB	11.3 g/dl	9.8–16.2
MCV	46.1 fl	35.9–53.1
MCH	15.3 pg	11.8–17.3
MCHC	33.2 g/dl	28.1–35.8
RDW	22.9%	15.0–27.0
%RETIC	1.0%	
RETIC	70.8 K/μl	3.0–50.0
RETIC-HGB	17.3 pg	13.2–20.8
WBC	25.24 x 10 ⁹ /l	2.87–17.02
%NEU	79.9%	
%LYM	10.7%	
%MONO	5.7%	
%EOS	3.6%	
%BASO	0.1%	
NEU	20.17 x 10 ⁹ /l	2.30–10.29
LYM	2.69 x 10 ⁹ /l	0.92–6.88
MONO	1.44 x 10 ⁹ /l	0.05–0.67
EOS	0.92 x 10 ⁹ /l	0.17–1.57
BASO	0.02 x 10 ⁹ /l	0.01–0.26
PLT	608 K/μl	151–600
MPV	15.6 fl	11.4–21.6
PCT	0.95%	0.17–0.86

BASO = Basophils; EOS = Eosinophils; HCT = hematocrit; HGB = Hemoglobin; LYM = Lymphocytes; MCH = mean cell hemoglobin; MCHC = mean cell hemoglobin concentration; MCV = mean cell volume; MONO = Monocytes; MPV = Mean Platelet Volume; NEU = Neutrophils; PCT = Procalcitonin; PLT = Platelet count; RBC = Red Blood cells; RDW = Red Cell Distribution width; RETIC = Reticulocyte; Retic-HGB = Reticulocyte Hemoglobin; WBC = White Blood Cells.

(Ambrosini *et al.*, 2021). It originates from DNA mutations during primary differentiation of secondary lymphoid tissues, with potential viral and environmental factors and possible genetic predisposition (Argyle and Pecceu, 2016).

Lymphomas are mainly categorized according to their topography (Sato Y Fujino and Chino, 2014). There are alimentary, mediastinal, multicentric, and extranodal lymphomas (Sato Y Fujino and Chino, 2014), where extranodal lymphoma refers to a disease confined to locations other than the alimentary, mediastinal, or multicentric sites, including the nasal, central nervous system, cutaneous, Ocular, laryngeal,

Table 2. Biochemistry when the cat was diagnosed with renal lymphoma. The only abnormalities were an increased urea and decreased ALKP.

Blood parameter	Value patient	Normal values
GLU	6.53 mmol/l	4.11–8.84
CREA	157 μmol/l	71–212
UREA	13.9 mmol/l	5.7–12.9
BUN/CREA	22	
PHOS	1.50 mmol/l	1.00–2.42
CA	2.42 mmol/l	1.95–2.83
TP	72 g/l	57–89
ALB	30 g/l	22–40
GLOB	42 g/l	28–51
ALB/GLOB	0.7	
ALT	47 U/l	12–130
ALKP	12 U/l	14–111
GGT	0 U/l	0–4
TBIL	11 μmol/l	0–15
CHOL	2.34 mmol/l	1.68–5.81
AMYL	1362 U/l	500–1500
LIPA	457 U/l	100–1400
Na	160 mmol/l	150–165
K	3.8 mmol/l	3.5–5.8
Na/K	42	
Cl	116 mmol/l	112–129
Osm Calc	325 mmol/kg	

ALB = Albumin; ALKP = Alkaline Phosphatase; ALT Alanine Transaminase; AMY = Amylase; BUN = Blood urea nitrogen; CA = Calcium; CHOL = Cholesterol; Cl = Chloride; CREA = creatinine; GGT = Gamma-glutamyl Transferase; GLOB = Globulin; GLU = glucose; K = Potassium; LIPA = Lipase; Na = Sodium; Osm Calc = Osmolarity Calcium; PHOS = phosphate; TBIL = total Bilirubin; TP = total protein; UREA = urea.

and renal lymphoma (Argyle and Pecceu, 2016). Renal lymphoma can be primary or associated with gastrointestinal or multicentric involvement (Ambrosini *et al.*, 2021). Clinical signs are widely variable and nonspecific because they are associated with the affected organ (Ambrosini *et al.*, 2021). Cats diagnosed with renal lymphoma often exhibit signs of renal disease, such as polyuria, polydipsia, azotemia, and nonregenerative anemia (Moore, 2013). Ultrasound can be used for the diagnosis of renal lymphoma and is also used to guide fine needle aspiration. Cytology is highly sensitive and specific for diagnosis (Moore *et al.*, 2013).

Neoplastic lymphocytes are large with relatively abundant cytoplasm and large nuclei that contain pale chromatin and one or more prominent nucleoli (Twomey and Alleman, 2005). Chemotherapy is the

Table 3. Urine analysis including Urine Protein to Creatinine ratio when the cat was diagnosed with renal lymphoma. Only proteinuria was found.

Test	Result	Reference range	Units
Physical examination			
Color	Light yellow	N/A	-
Turbidity	Clear	N/A	-
Specific gravity	1.041	N/A	-
Dipstick			
PH	6.5	6–7	-
Protein	2+ (1.0 g/l)	N/A	-
Glucose	Negative	Negative	-
Heme	Negative	Negative	-
Bilirubin	Negative	N/A	-
Ketones	Negative	Negative	-
Microscopic examination			
RBC	None seen	<5	/HPF
WBC	<5	<5	/HPF
Bacteria	None seen	N/A	/HPF
Epithelial cells	None seen	N/A	/HPF
Crystals	None seen	N/A	/LPF
Casts	None seen	N/A	/LPF
Others	Few-debris, few-fat drops	N/A	-
Biochemistry			
Urine protein/creatinine ratio (UPC)	0.39	<0.50	-

HPF = high power field; LPW = low power field; N/A = not available.

treatment of choice for large, high-grade lymphoma (Moore, 2013). The prognosis for large-cell/high-grade lymphoma mainly depends on the response to chemotherapy treatment and anatomical form. Nasal and mediastinal lymphoma carry the best prognosis, most likely due to the high response rate (Pui Yung Anna *et al.*, 2025).

Long-term follow-up of more than 2–3 years for patients treated for renal lymphoma is only rarely reported (Williams *et al.*, 2021). In this report, we present a case of FIP infection following chemotherapy in a young cat treated for renal lymphoma. The cat achieved a complete response to both FIP treatment and lymphoma, and he is currently free of both diseases 5 years after diagnosis.

Case details

A neutered 3.9-year-old domestic short-haired male cat was referred for further investigation of a large mass in the cranial abdomen. At presentation, a 2-week history of hyporexia and lethargy was reported. On physical examination, besides a huge palpable mass in the mid-cranial abdomen, occupying two-thirds of the abdomen, no other abnormalities were observed.

Hematology (Table 1), full biochemistry profile (Table 2), and urine protein to creatinine ratio (UPC) (Table 3) showed no significant abnormalities. Immunodeficiency virus and feline leukemia virus were both negative on the SNAP test (4Dx Plus; IDEXX Laboratories, Westbrook, ME). CT angiography of the thorax and abdomen revealed left renomegaly measuring approximately 80 mm in width and 65 mm in height and 90 mm in length with a markedly deformed structure, no corticomedullary definition, and a large lobulated expansile appearance (Fig. 1). The mass effect led to severe lateral intestine displacement into the right abdomen and craniolateral displacement of the spleen, which curled in the left cranial abdomen. The right kidney was mildly enlarged, approximately 50 mm in length, with several expansile, rounded lesions within the cortex and medulla. The rest of the CT examination was unremarkable. Ultrasound-guided fine needle angiography aspirates of the mass arising from the left kidney revealed a cell population of large-cell lymphoma (Fig. 2). A Cyclophosphamide, vincristine, prednisolone and vinblastine (COVP) protocol was initiated. This comprises a modified COP protocol with vinblastine addition (Pui Yung Anna, 2025). Complete remission was achieved after the

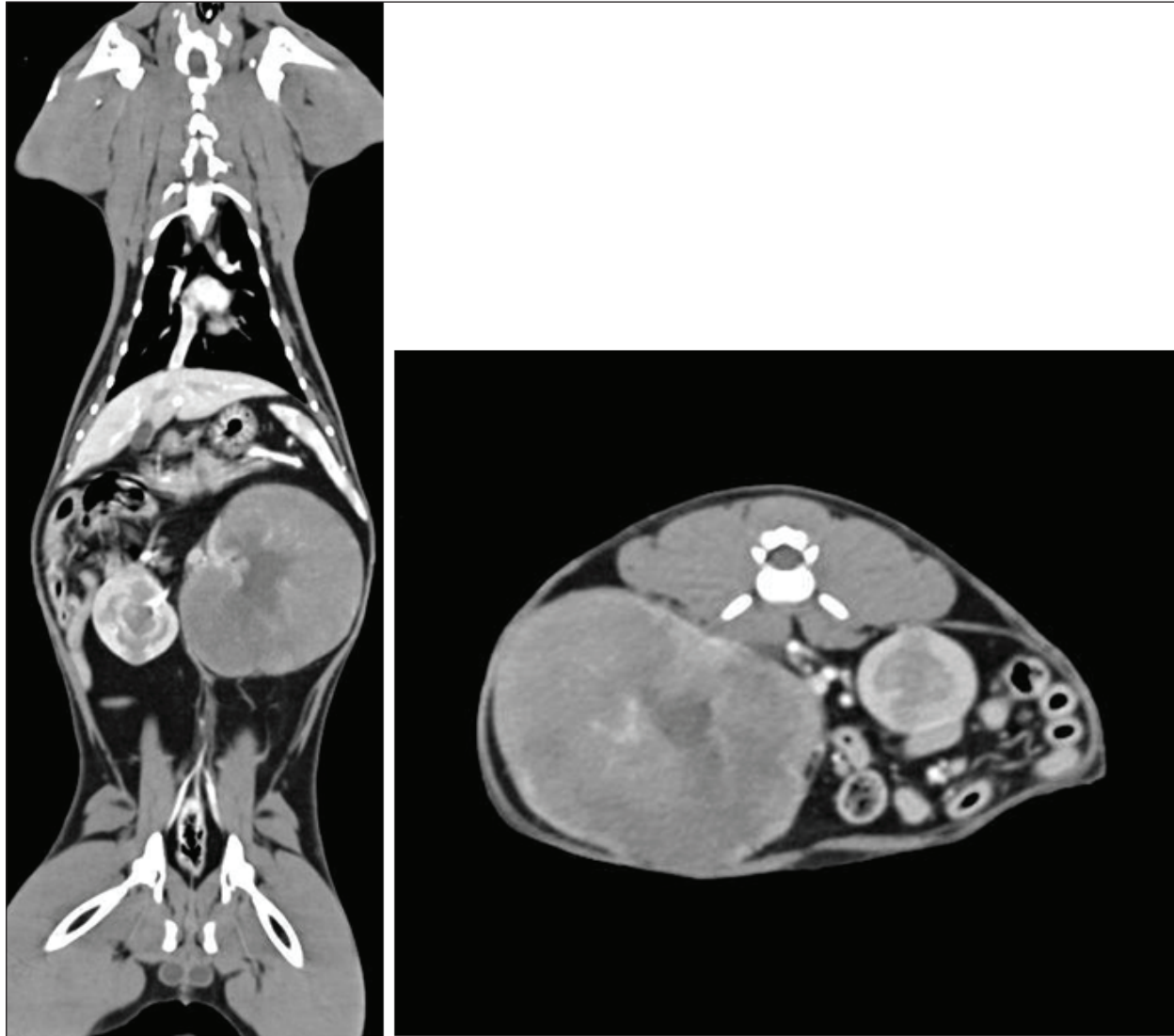


Fig. 1. Soft tissue window CT scan of the left coronal and right axis. The left kidney is markedly enlarged and measures approximately 80 mm in width, 65 mm in height, and 90 mm in length. It has a markedly deformed structure with no cortical medullary definition and a large lobulated expansile appearance. The cortex and medulla show reduced contrast enhancement, whereas the irregular peripheral rim of the tissue shows increased contrast enhancement. It retains a central area resembling a renal pelvis, where the left renal artery and vein are seen to enter; however, no accumulation of contrast medium is seen. The left kidney has a mass effect with severe lateral displacement of the intestines into the right abdomen. It also causes craniolateral displacement of the spleen, which is curled in the left cranial abdomen. The right kidney is mildly enlarged, measuring approximately 5 cm in length. It has several expansile rounded lesions within the cortex and medulla that mildly deform the external contour of the kidney. They are similar in attenuation to the medulla precontrast (approximately 40 HU) but show markedly reduced contrast enhancement.

second week of the COVP protocol. No significant AEs attributed to chemotherapy were reported during the chemotherapy course.

Twenty weeks after the start of chemotherapy, during the maintenance injection protocol of vincristine and cyclophosphamide, anorexia and lethargy were reported by the patient.

Examination revealed moderate dyspnea with paradoxical breathing and a fever of 40°C. Bloodwork revealed mild nonregenerative anemia 25.3% (30.3–52.3), neutrophilia $22.35 \times 10^9/l$ (2.30–10.29), and mild hyperglobulinemia 54 g/l (28–51) (Table 4).

Thoracic radiographs revealed a large amount of pleural effusion. Therapeutic thoracocentesis was performed,

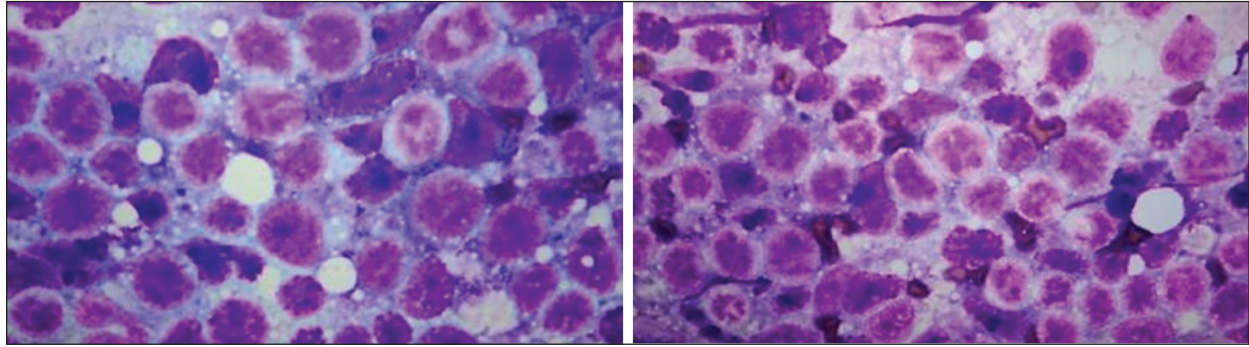


Fig. 2. 100x magnification of the fine-needle aspirate originating from the left kidney mass. The sample is highly cellular and contains large lymphocytes, 2-3 times the size of a neutrophil. The lymphoid population is dominated by large lymphocytes with scant dark blue cytoplasm. A few punctate clear cytoplasmic vacuoles were present in some cells. The nuclei are round and have fine reticular chromatin and one or more prominent nucleoli. No normal renal tissue is seen. No evidence of inflammation or infection. The finding of a predominance of larger lymphocytes in the tissue is consistent with the findings of large cell lymphoma.

and the fluid was submitted for evaluation. Fluid analysis showed an exudate with a high protein count, and a proteinaceous background with neutrophilic and macrophage inflammation; negative aerobic culture for bacteria (Fig. 3). A PCR test for feline coronavirus was performed on the fluid, and a positive result was obtained (Table 5).

Echocardiography and abdominal ultrasonography performed at this point revealed no signs of cardiac arrest. Cardiomyopathy that could have led to CHF or other abnormalities suggestive of lymphoma recurrence. A combination of blood tests, fluid analysis, and exclusion of other abnormalities using ultrasound confirmed the suspicion of FIP.

Dexamethasone administered intravenously at 0.2 mg/kg q 24 hours was started, while the FeCoV PCR test result was pending. After a short initial clinical improvement, pleural effusion reoccurred 2 days later, and therapeutic thoracocentesis was repeated. Following the confirmation of FIP by PCR, daily subcutaneous (SQ) injections of GS441524 were administered at a dose of 10 mg/kg for 12 weeks. Mild discomfort and irritation during SQ injection were the only side effects reported by the owner. Dexamethasone treatment was stopped once FIP was confirmed, and nucleoside analogs were started.

Marked clinical improvement was noticed after GS441524 was started with no clinical or abnormalities detected at 1–3–6–12 months rechecks. At the time this manuscript was written almost 5 years after the initial diagnosis of large-cell renal lymphoma, the cat is doing well without any clinical signs of lymphoma or recurrence of FIP.

Ethical approval

This study involved the use of client-owned animals only, following the internationally recognized high standards of individual clinical veterinary patient care. Ethical approval from a committee was not required.

Discussion

Lymphoma is the most common cancer affecting domestic cats, with an average age of 11 years (Pedersen *et al.*, 2019). Pediatric or juvenile lymphoma is rare in cats; however, it has been reported (Williams *et al.*, 2021; Rogato and Tanis, 2023). Recently, young cats with lymphoma tended to have mediastinal or disseminated forms, with only 15% renal origin (Rogato and Tanis, 2023). The incidence of renal lymphoma was 3.6% of all lymphoma cases in one study, where patients as young as 1-year-old were also reported to be affected (Williams *et al.*, 2021). Although age difference is not assessed, it is always a prognostic factor; cats with renal lymphoma treated with chemotherapy have a median survival of 203 days, with 3 of 27 cats surviving longer than 2 years (Williams *et al.*, 2021). This means that some cats could be cured of renal lymphoma. Unfortunately, these long-surviving cats were not reported, making it impossible to assess the relationship between young age and survival. Nevertheless, in one large study using CHOP-based chemotherapy, the anatomical form of lymphoma, approximately 30% of cats that achieved a complete response reported to live for 2 or more years; however, renal lymphoma tended to have the shortest survival compared with other forms (Collette *et al.*, 2016). The cat reported here is still alive, 4 years after renal lymphoma diagnosis without a reported relapse. As said by Williams *et al.* (2021) and Collette *et al.* (2016), a cure is a possibility in cases that achieve a long-term, durable complete response. However, retrospective case reports have a limitation, and there is a need for larger cohort studies.

Although multiagent chemotherapy is the therapy of choice for renal lymphoma, comparing the most commonly used chemotherapy protocols, such as COP (vincristine, cyclophosphamide, and prednisolone) and CHOP (vincristine, cyclophosphamide, and doxorubicin), is rare. Nonetheless, it appears that

Table 4. Bloodwork from cats when FIP was suspected. Nonregenerative anemia, neutrophilia with band neutrophils and hyperglobulinemia .

Blood parameter	Value patient	Normal values
RBC	$6.03 \times 10^{12}/l$	6.54–12.20
HCT	25.3%	30.3–52.3
HGB	8.5 g/dl	9.8–16.2
MCV	42.0 fl	35.9–53.1
MCH	14.1 pg	11.8–17.3
MCHC	33.6 g/dl	28.1–35.8
RDW	27.1%	15.0–27.0
%RETIC	0.5%	
RETIC	28.3 K/ μ l	3.0–50.0
RETIC-HGB	15.6 pg	13.2–20.8
WBC	$24.84 \times 10^9/l$	2.87–17.02
%NEU	*89.9%	
%LYM	*8.0%	
%MONO	*1.8%	
%EOS	0.2%	
%BASO	0.1%	
NEU	$22.35 \times 10^9/l$	2.30–10.29
BAND	*Suspected	
LYM	$*8.00 \times 10^9/l$	0.92–6.88
MONO	$*1.98 \times 10^9/l$	0.05–0.67
EOS	$*0.44 \times 10^9/l$	0.17–1.57
BASO	$0.05 \times 10^9/l$	0.01–0.26
PLT	427 K/ μ l	151–600
MPV	16.5 fl	11.4–21.6
PCT	0.70%	0.17–0.86
GLU	8.42 mmol/l	4.11–8.84
CREA	135 μ mol/l	71–212
UREA	5.7 mmol/l	5.7–12.9
BUN/CREA	10	
TP	80 g/l	57–89
ALB	26 g/l	22–40
GLOB	54 g/l	28–51
ALB/GLOB	0.5	
ALT	61 U/l	12–130
ALKP	<10 U/l	14–111

cats that achieve complete response tend to live for a long time, independent of the protocol used (Bernardo Marques, F. Ponce and Beaudu-Lange, 2024). In this case, a modified COP protocol with vinblastine addition (COVP protocol) was used. This is routinely

used by the clinic of the lead author (Pui Yung Anna *et al.*, 2025).

Feline infectious peritonitis is an immune-mediated disease secondary to systemic infection of macrophages by a mutant feline coronavirus strain (Thayer *et al.*, 2022). Before the use of nucleoside analogs such as remdesivir (GS5734) or GS441524, a median survival of 9 days was expected in cats with FIP (Ritz *et al.*, 2007). Treatment with GS441524 proved to be effective against FIP in both the short and long term, with no confirmed relapses during a 12-month follow-up period and a 96% survival rate at 6 months in cats that survived beyond the first 48 hours (Coggins *et al.*, 2023; Zwicklbauer *et al.*, 2023). In this report, SQ GS441524 was used. Although very effective, daily SQ injections of nucleoside analogs are associated with pain, vocalization, and injection site reactions, making their administration challenging at times, as reported in our case (Green *et al.*, 2023; Taylor *et al.*, 2023; Negash *et al.*, 2024). SQ GS441524 administration in the present study case was chosen due to the lack of availability of both oral form and information at the time of the efficacy of oral GS441524. Oral administration is now widely available and is the treatment of choice used by the authors, as shown by long-term follow-up of cat remissions (Krentz *et al.*, 2021; Zwicklbauer *et al.*, 2023).

Stressful events during a young age, including surgical procedures such as spaying and neutering, have been associated with FIP development (Pedersen, 2009). Stress suppresses the immune system and increases the risk of FIP via a higher rate of viral replication mutation (Riemer *et al.*, 2016). Long-term chemotherapy can contribute to immune defense, but further research is needed to confirm its correlation and predisposition for FIP.

Following recovery from immunosuppression, immune reconstitution might have contributed to the following: the development of FIP. This is known to occur because of a change in the T helper responses to inflammation after discontinuation of immunosuppressive drugs (Sueki *et al.*, 2019).

Positive FCoV result from ascites fluid has been anecdotally reported in a young cat following chemotherapy for jejunal lymphoblastic lymphosarcoma (Borku *et al.*, 2017).

The specificity of the RT RNA PCR test on effusion is 95.8%, and the sensitivity is 68.6% (Felten *et al.*, 2017). Hematological abnormalities are very common in cats with FIP. The following may be seen but may not always be present: nonregenerative anemia, microcytosis, lymphopenia, neutrophilia, thrombocytopenia, hyperproteinemia, hyperglobulinemia, hypoalbuminemia, hyperbilirubinemia, azotemia, and increased liver enzyme activity [Felten and Hartmann (2019)]. The albumin-to-globulin ratio can be used to predict FIP diagnosis. Several cutoffs have been made

Table 5. Examination results from the fluid collected from the thorax. The fluid is an exudate, and the PCR test result for FIP is positive, which diagnoses the cat.

Test	Specimen	Specimen Site	Result	Reference range
<i>Feline Coronavirus</i>	Fluid	Thorax	Positive	Not applicable
Thoracic fluid, cat	Result	Reference Interval		
Gross appearance	Light yellow, slightly cloudy	clear		
RBCC	1.04	0 x 10 ⁹ /l		
Nucleated cell count	8.44	<1.5 x 10 ⁹ /l		
Total protein	58	<20 g/l		

to rule in (<0.4) or rule out (>0.6–0.8) FIP (Felten and Hartmann, 2019).

The development of a debilitating condition, such as renal lymphoma and immunosuppression caused by chemotherapeutic agents, might have played a role in the FIP development in this case.

To the authors' knowledge, this is the first report documenting the development of FIP following the successful treatment of renal lymphoma. Regardless of the reason for the two potentially fatal pathologies, the current case report highlights the need for effective therapeutic options to be available and long-term remissions, or even a potential cure, are possible if appropriate therapies and care.

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Authors' contributions

I.B. and A.G. produced the original draft of the manuscript. A.A., A.G., and I.B. revised the manuscript. All authors have read and approved the final version of the manuscript.

Conflict of interest

The author declares that there is no conflict of interest.

Data availability

All data supporting this study's findings are available within the manuscript.

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